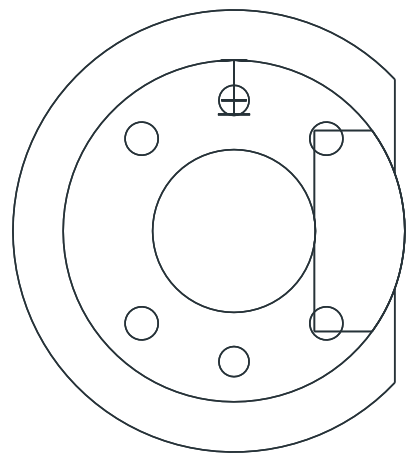


MFG-EX-BAR-01 · v0.8 MANUFACTURING DRAWING

TOP X-Y



FRONT X-Z



SIDE Y-Z



Drawing MFG-EX-BAR-01 | Part EX-BAR-01 — ID16.20 x OD34 barrel | Qty 1

Rev final-design-fabrication-closure-v0.8 | Units mm | Scale NTS, dimensions govern | Third-angle projection | source commit 93ec6a9c0c2afa3ad72288a967f07b7c8d03c99e

Material: SCM440 (KS D3867/JIS G4105 equivalent) QT + gas nitride bore | Process/finish: machine integral rear shoulder from oversized blank, deep drill, stress relieve, ream/hone, port, die-interface thread and sensor bores, nitride, final hone

Overall: X 39.000 × Y 44.000 × Z 280.000 mm | general tolerance ISO 2768-m unless noted

Critical: L280; ID16.20 +0.02/0 honed; OD34 -0.03/0 ground; rear shoulder Ø44 -0.05/0 x8 +/-0.02, feed flat X16; face runout0.03 to Datum D; 4xM4-6H depth8 PCD26; 2xØ3 H7 locating holes PCD26 at0/180 deg, position Ø0.02 to D, selected Ø3 m6 dowels; 3xØ3.20 +0.05/0 flat-bottom blind5.40 +/-0.05 at B+95/170/245; 2xM3-6H depth4 at axial pitch10 around each sensor; sensor position0.05 to D, perpendicularity0.10/5.5; conservative ligament3.345 >=3.32; thread ligament outer/inner>=2.0/2.9; B rear face/C front face/D bore axis; HOLD material/process; SCM440 QT 28–32 HRC; gas nitride 0.30–0.50 mm, surface ≥900 HV; final hone bore Ra 0.4–0.8 µm; three-station bore report

GD&T/datums: B=rear face; C=front face; D=bore axis; nominal +X=feed, +Y=sensors; die holes 45+90n deg from +X; angle tolerance HOLD; verify datum marking on supplier inspection report

Inspection: CMM/caliper against STEP and drawing; deburr and visual crack/warp inspection; certified material/heat-treatment report; three-station bore/OD and TIR report

Edges: remove burrs; break unspecified sharp edges C0.3–0.5. No unapproved material/tolerance substitution.

Artifact check: PASS only; fabrication HOLD pending full v0.8 gates. Physical NOT_RUN; machining USER_APPROVAL_REQUIRED.